

Table 1: Correction payloads, paths, and tags used in the analysis.

Era	Correction	Payload / path	Key / tag
Run3_2022	JEC/JER	/cvmfs/cms-griddata.cern.ch/cat/metadata/JME/Run3-22CDSep23-Summer22-NanoA0Dv12/latest/jet_jerc.json.gz	JEC MC: Summer22_22Sep2023_V4_MC; JEC data: Summer22_22Sep2023_V4_DATA; JER: Summer22_22Sep2023_JRV2_MC
	JER smearing	/cvmfs/cms-griddata.cern.ch/cat/metadata/JME/JER-Smearing/latest/jer_smear.json.gz	sources: JER, Total
	Pileup	/cvmfs/cms-griddata.cern.ch/cat/metadata/LUM/Run3-22CDSep23-Summer22-NanoA0Dv12/latest/puWeights.json.gz	Collisions2022_355100_357900_eraBCD_GoldenJson
	Muon SF	/cvmfs/cms-griddata.cern.ch/cat/metadata/MUO/Run3-22CDSep23-Summer22-NanoA0Dv12/latest/muon_Z.json.gz	NUM_LooseID_DEN_TrackerMuons, NUM_MediumID_DEN_TrackerMuons, NUM_TightID_DEN_TrackerMuons, NUM_LoosePFIso_DEN_LooseID, NUM_LoosePFIso_DEN_MediumID, NUM_LoosePFIso_DEN_TightID, NUM_TightPFIso_DEN_TightID, NUM_LooseMiniIso_DEN_LooseID, NUM_LooseMiniIso_DEN_MediumID, NUM_IsoMu24_DEN_CutBasedIdMedium_and_PFIsoMedium, NUM_IsoMu24_DEN_CutBasedIdTight_and_PFIsoTight
	Muon ScaRe	corrections/data/MUO/MuonScaRe/Run3-22CDSep23-Summer22-NanoA0Dv12/muon_scalesmearing.json	standard; VXBS payload: corrections/data/MUO/MuonScaRe/Run3-22CDSep23-Summer22-NanoA0Dv12/muon_scalesmearing_VXBS.json
	Muon FSR	corrections/muon_fsr.py	algorithmic; no payload
	Jet veto map	/cvmfs/cms-griddata.cern.ch/cat/metadata/JME/Run3-22CDSep23-Summer22-NanoA0Dv12/latest/jetvetomaps.json.gz	Summer22_23Sep2023_RunCD_V1
	b-tag WP	/cvmfs/cms-griddata.cern.ch/cat/metadata/BTV/Run3-22CDSep23-Summer22-NanoA0Dv12/latest/btagging.json.gz	PNet_wp_values (L, M, T)
Run3_2022EE	Golden JSON	corrections/data/golden_json/Cert_Collisions2022_355100_362760_Golden.json	data luminosity mask
	Trigger SF	HLT_IsoMu24	NUM_IsoMu24_DEN_CutBasedIdMedium_and_PFIsoMedium
	JEC/JER	/cvmfs/cms-griddata.cern.ch/cat/metadata/JME/Run3-22EFGSep23-Summer22EE-NanoA0Dv12/latest/jet_jerc.json.gz	JEC MC: Summer22EE_22Sep2023_V4_MC; JEC data: Summer22EE_22Sep2023_Run{}_V4_DATA, Summer22EE_22Sep2023_V4_DATA; JER: Summer22EE_22Sep2023_JRV2_MC
	JER smearing	/cvmfs/cms-griddata.cern.ch/cat/metadata/JME/JER-Smearing/latest/jer_smear.json.gz	sources: JER, Total
	Pileup	/cvmfs/cms-griddata.cern.ch/cat/metadata/LUM/Run3-22EFGSep23-Summer22EE-NanoA0Dv12/latest/puWeights.json.gz	Collisions2022_359022_362760_eraEFG_GoldenJson
	Muon SF	/cvmfs/cms-griddata.cern.ch/cat/metadata/MUO/Run3-22EFGSep23-Summer22EE-NanoA0Dv12/latest/muon_Z.json.gz	NUM_LooseID_DEN_TrackerMuons, NUM_MediumID_DEN_TrackerMuons, NUM_TightID_DEN_TrackerMuons, NUM_LoosePFIso_DEN_LooseID, NUM_LoosePFIso_DEN_MediumID, NUM_LoosePFIso_DEN_TightID, NUM_TightPFIso_DEN_TightID, NUM_LooseMiniIso_DEN_LooseID, NUM_LooseMiniIso_DEN_MediumID, NUM_IsoMu24_DEN_CutBasedIdMedium_and_PFIsoMedium, NUM_IsoMu24_DEN_CutBasedIdTight_and_PFIsoTight
	Muon ScaRe	corrections/data/MUO/MuonScaRe/Run3-22EFGSep23-Summer22EE-NanoA0Dv12/muon_scalesmearing.json	standard; VXBS payload: corrections/data/MUO/MuonScaRe/Run3-22EFGSep23-Summer22EE-NanoA0Dv12/muon_scalesmearing_VXBS.json
	Muon FSR	corrections/muon_fsr.py	algorithmic; no payload
Run3_2023	Jet veto map	/cvmfs/cms-griddata.cern.ch/cat/metadata/JME/Run3-22EFGSep23-Summer22EE-NanoA0Dv12/latest/jetvetomaps.json.gz	Summer22EE_23Sep2023_RunEFG_V1
	b-tag WP	/cvmfs/cms-griddata.cern.ch/cat/metadata/BTV/Run3-22EFGSep23-Summer22EE-NanoA0Dv12/latest/btagging.json.gz	PNet_wp_values (L, M, T)
	Golden JSON	corrections/data/golden_json/Cert_Collisions2022_355100_362760_Golden.json	data luminosity mask
	Trigger SF	HLT_IsoMu24	NUM_IsoMu24_DEN_CutBasedIdMedium_and_PFIsoMedium
	JEC/JER	/cvmfs/cms-griddata.cern.ch/cat/metadata/JME/Run3-23CSep23-Summer23-NanoA0Dv12/latest/jet_jerc.json.gz	JEC MC: Summer23Prompt23_V4_MC; JEC data: Summer23Prompt23_Run{}_V4_DATA, Summer23Prompt23_V4_DATA; JER: Summer23Prompt23_RunCv1234_JRV3_MC
	JER smearing	/cvmfs/cms-griddata.cern.ch/cat/metadata/JME/JER-Smearing/latest/jer_smear.json.gz	sources: JER, Total
	Pileup	/cvmfs/cms-griddata.cern.ch/cat/metadata/LUM/Run3-23CSep23-Summer23-NanoA0Dv12/latest/puWeights.json.gz	Collisions2023_366403_369802_eraBC_GoldenJson
	Muon SF	/cvmfs/cms-griddata.cern.ch/cat/metadata/MUO/Run3-23CSep23-Summer23-NanoA0Dv12/latest/muon_Z.json.gz	NUM_LooseID_DEN_TrackerMuons, NUM_MediumID_DEN_TrackerMuons, NUM_TightID_DEN_TrackerMuons, NUM_LoosePFIso_DEN_LooseID, NUM_LoosePFIso_DEN_MediumID, NUM_LoosePFIso_DEN_TightID, NUM_TightPFIso_DEN_TightID, NUM_LooseMiniIso_DEN_LooseID, NUM_LooseMiniIso_DEN_MediumID, NUM_IsoMu24_DEN_CutBasedIdMedium_and_PFIsoMedium, NUM_IsoMu24_DEN_CutBasedIdTight_and_PFIsoTight
Run3_2023BPix	Muon ScaRe	corrections/data/MUO/MuonScaRe/Run3-23CSep23-Summer23-NanoA0Dv12/muon_scalesmearing.json	standard; VXBS payload: corrections/data/MUO/MuonScaRe/Run3-23CSep23-Summer23-NanoA0Dv12/muon_scalesmearing_VXBS.json
	Muon FSR	corrections/muon_fsr.py	algorithmic; no payload
	Jet veto map	/cvmfs/cms-griddata.cern.ch/cat/metadata/JME/Run3-23CSep23-Summer23-NanoA0Dv12/latest/jetvetomaps.json.gz	Summer23Prompt23_RunC_V1
	b-tag WP	/cvmfs/cms-griddata.cern.ch/cat/metadata/BTV/Run3-23CSep23-Summer23-NanoA0Dv12/latest/btagging.json.gz	PNet_wp_values (L, M, T)
	Golden JSON	corrections/data/golden_json/Cert_Collisions2023_366442_370790_Golden.json	data luminosity mask
	Trigger SF	HLT_IsoMu24	NUM_IsoMu24_DEN_CutBasedIdMedium_and_PFIsoMedium
	JEC/JER	/cvmfs/cms-griddata.cern.ch/cat/metadata/JME/Run3-23DSep23-Summer23BPix-NanoA0Dv12/latest/jet_jerc.json.gz	JEC MC: Summer23BPixPrompt23_V4_MC; JEC data: Summer23BPixPrompt23_Run{}_V4_DATA, Summer23BPixPrompt23_V4_DATA; JER: Summer23BPixPrompt23_RunD_JRV3_MC
	JER smearing	/cvmfs/cms-griddata.cern.ch/cat/metadata/JME/JER-Smearing/latest/jer_smear.json.gz	sources: JER, Total
Run3_2023BPix	Pileup	/cvmfs/cms-griddata.cern.ch/cat/metadata/LUM/Run3-23DSep23-Summer23BPix-NanoA0Dv12/latest/puWeights.json.gz	Collisions2023_369803_370790_eraD_GoldenJson
	Muon SF	/cvmfs/cms-griddata.cern.ch/cat/metadata/MUO/Run3-23DSep23-Summer23BPix-NanoA0Dv12/latest/muon_Z.json.gz	NUM_LooseID_DEN_TrackerMuons, NUM_MediumID_DEN_TrackerMuons, NUM_TightID_DEN_TrackerMuons, NUM_LoosePFIso_DEN_LooseID, NUM_LoosePFIso_DEN_MediumID, NUM_LoosePFIso_DEN_TightID, NUM_TightPFIso_DEN_TightID, NUM_LooseMiniIso_DEN_LooseID, NUM_LooseMiniIso_DEN_MediumID, NUM_IsoMu24_DEN_CutBasedIdMedium_and_PFIsoMedium, NUM_IsoMu24_DEN_CutBasedIdTight_and_PFIsoTight
	Muon ScaRe	corrections/data/MUO/MuonScaRe/Run3-23DSep23-Summer23BPix-NanoA0Dv12/muon_scalesmearing.json	standard; VXBS payload: corrections/data/MUO/MuonScaRe/Run3-23DSep23-Summer23BPix-NanoA0Dv12/muon_scalesmearing_VXBS.json
	Muon FSR	corrections/muon_fsr.py	algorithmic; no payload

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Table 1 – continued

Era	Correction	Payload / path	Key / tag
Run3_2024	Jet veto map	/cvmfs/cms-griddata.cern.ch/cat/metadata/JME/Run3-23DSep23-Summer23BPix-NanoAODv12/latest/jetvetomaps.json.gz	Summer23BPixPrompt23_RunD_V1
	b-tag WP	/cvmfs/cms-griddata.cern.ch/cat/metadata/BTV/Run3-23DSep23-Summer23BPix-NanoAODv12/latest/btagging.json.gz	PNet_wp_values (L, M, T)
	Golden JSON	corrections/data/golden_json/Cert_Collisions2023_366442_370790_Golden.json	data luminosity mask
	Trigger SF	HLT_IsoMu24	NUM_IsoMu24_DEN_CutBasedIdMedium_and_PFIsoMedium
	JEC/JER	/cvmfs/cms-griddata.cern.ch/cat/metadata/JME/Run3-24CDEReprocessingFGHIPrompt-Summer24-NanoAODv15/latest/jet_jerc.json.gz	JEC MC: Summer24Prompt24_V5_MC; JEC data: Summer24Prompt24_V5_DATA; JER: Summer24Prompt24_JRV2_MC
	JER smearing	/cvmfs/cms-griddata.cern.ch/cat/metadata/JME/JER-Smearing/latest/jer_smear.json.gz	sources: JER, Total
	Pileup	/cvmfs/cms-griddata.cern.ch/cat/metadata/LUM/Run3-24CDEReprocessingFGHIPrompt-Summer24-NanoAODv15/latest/puWeights_BCDEFGHI.json.gz	Collisions24_BCDEFGHI_goldenJSON
	Muon SF	/cvmfs/cms-griddata.cern.ch/cat/metadata/MUO/Run3-24CDEReprocessingFGHIPrompt-Summer24-NanoAODv15/latest/muon_Z.json.gz	NUM_LooseID_DEN_TrackerMuons, NUM_MediumID_DEN_TrackerMuons, NUM_TightID_DEN_TrackerMuons, NUM_LoosePFIso_DEN_LooseID, NUM_LoosePFIso_DEN_MediumID, NUM_LoosePFIso_DEN_TightID, NUM_TightPFIso_DEN_TightID, NUM_IsoMu24_DEN_CutBasedIdMedium_and_PFIsoMedium, NUM_IsoMu24_DEN_CutBasedIdTight_and_PFIsoTight
	Muon ScaRe	corrections/data/MUO/MuonScaRe/Run3-24CDEReprocessingFGHIPrompt-Summer24-NanoAODv15/muon_scalesmearing.json	standard; VXBS payload: corrections/data/MUO/MuonScaRe/Run3-24CDEReprocessingFGHIPrompt-Summer24-NanoAODv15/muon_scalesmearing_VXBS.json
	Muon FSR	corrections/muon_fsr.py	algorithmic; no payload
Run3_2025	Jet veto map	/cvmfs/cms-griddata.cern.ch/cat/metadata/JME/Run3-24CDEReprocessingFGHIPrompt-Summer24-NanoAODv15/latest/jetvetomaps.json.gz	Summer24Prompt24_RunBCDEFGHI_V1
	b-tag WP	/cvmfs/cms-griddata.cern.ch/cat/metadata/BTV/Run3-24CDEReprocessingFGHIPrompt-Summer24-NanoAODv15/latest/btagging.json.gz	UParTAK4_wp_values (L, M, T)
	Golden JSON	corrections/data/golden_json/Cert_Collisions2024_378981_386951_Golden.json	data luminosity mask
	Trigger SF	HLT_IsoMu24	NUM_IsoMu24_DEN_CutBasedIdMedium_and_PFIsoMedium
	JEC/JER	/cvmfs/cms-griddata.cern.ch/cat/metadata/JME/Run3-25Prompt-Summer24-NanoAODv15/latest/jet_jerc.json.gz	JEC MC: Summer24Prompt25_V3_MC; JEC data: Summer24Prompt25_V3_DATA; JER: Summer24Prompt25_JRV2_MC
	JER smearing	/cvmfs/cms-griddata.cern.ch/cat/metadata/JME/JER-Smearing/latest/jer_smear.json.gz	sources: JER, Total
	Pileup	/cvmfs/cms-griddata.cern.ch/cat/metadata/LUM/Run3-25Prompt-Summer24-NanoAODv15/latest/puWeights_2025pp_Golden_Summer24_25ns_69200ub.json.gz	Collisions25_goldenJSON
	Muon SF	/afs/cern.ch/work/v/vdamante/Hmm_newSkin/corrections/data/MUO/SF/Run3-25Prompt-Summer24-NanoAODv15/muon_Z.json.gz	NUM_LooseID_DEN_TrackerMuons, NUM_MediumID_DEN_TrackerMuons, NUM_TightID_DEN_TrackerMuons, NUM_LoosePFIso_DEN_LooseID, NUM_LoosePFIso_DEN_MediumID, NUM_LoosePFIso_DEN_TightID, NUM_TightPFIso_DEN_TightID, NUM_LooseMiniIso_DEN_LooseID, NUM_LooseMiniIso_DEN_MediumID, NUM_IsoMu24_DEN_CutBasedIdMedium_and_PFIsoMedium, NUM_IsoMu24_DEN_CutBasedIdTight_and_PFIsoTight
	Muon ScaRe	corrections/data/MUO/MuonScaRe/Run3-25Prompt-Summer24-NanoAODv15/muon_scalesmearing.json	standard; VXBS payload: corrections/data/MUO/MuonScaRe/Run3-25Prompt-Summer24-NanoAODv15/muon_scalesmearing_VXBS.json
	Muon FSR	corrections/muon_fsr.py	algorithmic; no payload
Run3_2026	Jet veto map	/cvmfs/cms-griddata.cern.ch/cat/metadata/JME/Run3-25Prompt-Summer24-NanoAODv15/latest/jetvetomaps.json.gz	Summer24Prompt25_RunCDEFG_V1
	b-tag WP	/cvmfs/cms-griddata.cern.ch/cat/metadata/BTV/Run3-25Prompt-Summer24-NanoAODv15/latest/btagging.json.gz	UParTAK4_wp_values (L, M, T)
	Golden JSON	corrections/data/golden_json/Cert_Collisions2025_391658_398903_Golden.json	data luminosity mask
	Trigger SF	HLT_IsoMu24	NUM_IsoMu24_DEN_CutBasedIdMedium_and_PFIsoMedium
	JEC/JER	/cvmfs/cms-griddata.cern.ch/cat/metadata/JME/Run3-26Prompt-Summer24-NanoAODv15/latest/jet_jerc.json.gz	JEC MC: Summer24Prompt26_V2_MC; JEC data: Summer24Prompt26_Run{}_V1_DATA, Summer24Prompt26_V1_DATA; JER: Summer24Prompt25_JRV2_MC
	JER smearing	/cvmfs/cms-griddata.cern.ch/cat/metadata/JME/JER-Smearing/latest/jer_smear.json.gz	sources: JER, Total
	Pileup	/cvmfs/cms-griddata.cern.ch/cat/metadata/LUM/Run3-25Prompt-Summer24-NanoAODv15/latest/puWeights_2025pp_Golden_Summer24_25ns_69200ub.json.gz	Collisions25_goldenJSON
	Muon SF	/cvmfs/cms-griddata.cern.ch/cat/metadata/MUO/Run3-26Prompt-Summer24-NanoAODv15/latest/muon_Z.json.gz	
	Muon ScaRe	corrections/data/MUO/MuonScaRe/Run3-26Prompt-Summer24-NanoAODv15/muon_scalesmearing.json	standard; VXBS payload: corrections/data/MUO/MuonScaRe/Run3-26Prompt-Summer24-NanoAODv15/muon_scalesmearing_VXBS.json
	Muon FSR	corrections/muon_fsr.py	algorithmic; no payload
Run3_2026	Jet veto map	/cvmfs/cms-griddata.cern.ch/cat/metadata/JME/Run3-26Prompt-Summer24-NanoAODv15/latest/jetvetomaps.json.gz	Summer24Prompt26_RunBCD_V1
	b-tag WP	/cvmfs/cms-griddata.cern.ch/cat/metadata/BTV/Run3-26Prompt-Summer24-NanoAODv15/latest/btagging.json.gz	UParTAK4_wp_values (L, M, T)
	Golden JSON	corrections/data/golden_json/Cert_Collisions2026_401624_403937_golden.json	data luminosity mask
	Trigger SF	HLT_IsoMu24	NUM_IsoMu24_DEN_CutBasedIdMedium_and_PFIsoMedium